Model Analysis Report

Storm Intensity Model

MKM-SI-001

Report Date: 27 March 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

Contents

1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Storm intensity at return periods by Pareto tail index (base mean=40, std=12, threshold=65).

α (tail index)	10-yr intensity	25-yr intensity	50-yr intensity	100-yr intensity
1.50	100	100	100	100
2.00	100	100	100	100
2.50	100	100	100	100
3.00	100	100	100	100

Table 2: Storm intensity at return periods by base mean ($\alpha = 2.0$, std=12, threshold=65).

Base mean	10-yr intensity	25-yr intensity	50-yr intensity	100-yr intensity
30	60.9	64.5	86.5	100
35	72.4	100	100	100
40	100	100	100	100
45	100	100	100	100
50	100	100	100	100

Table 3: Exceedance probability by storm intensity and tail index.

Intensity	$\alpha = 1.5$	$\alpha = 2.0$	$\alpha = 2.5$	$\alpha = 3.0$
40	0.5000	0.5000	0.5000	0.5000
50	0.2023	0.2023	0.2023	0.2023
60	0.0478	0.0478	0.0478	0.0478
70	0.0167	0.0160	0.0155	0.0149
80	0.0136	0.0123	0.0111	0.0100
90	0.0114	0.0097	0.0083	0.0070

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 4: Stress testing scenarios for Storm Intensity Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	$+2\sigma$ inputs	Medium	Pending
Severe stress	$+3\sigma$ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 5: Performance metrics for Storm Intensity Model.

Metric	**Value**	**Status**
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.